



E-COMMERCE SALES ANALYSIS & BUSINESS INTELLIGENCE DASHBOARD

End-to-end Data Analysis Project using
SQL and Power BI

ABSTRACT

This project focuses on analysing E-commerce sales data to uncover key business insights and support data-driven decision-making. The analysis includes data cleaning, transformation, and modelling using SQL, followed by interactive dashboard development in Power BI.

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E-Commerce Sales Analysis & Business Intelligence Dashboard

Overview

This project focuses on analysing E-commerce Sales data to identify key business trends, customer behaviour, and product performance.

The dataset covers sales transactions from 2014 to 2017, enabling analysis of trends over multiple years.

The analysis involves data cleaning, transformation, and modelling using SQL, followed by interactive dashboard creation in power BI to support data-driven decision-making.

Business Problem

E-commerce businesses generate large volumes of transactional data, but with proper analysis, it is difficult to extract meaningful insights.

The key challenge is to understand:

- Which products generate the most revenue
- Which customers contribute the most to sales
- Which regions perform well or underperform
- How sales trends change over time

This project aims to solve these challenges using structured data analysis and visualization.

Dataset Information

Source: Public Dataset (Sample Superstore / Kaggle)

The dataset contains transactional sales data including customer details, product information, order details, and regional data.

Time Period: 2014 - 2017

Total Records: 9994 rows

Total Columns: 21 features

The dataset contains sales-related information such as

- Order Date
- Shift Date
- Customer Name
- Country
- City
- State
- Region
- Product Name
- Category
- Sub-Category
- Quantity
- Discount
- Profit

Note: Only the most relevant features are highlighted for analysis, while the dataset contains additional supporting columns.

Project Structure

- 1_Raw_Data - Original dataset (Sample Superstore)
- 2_SQL_Scripts - SQL scripts for data cleaning, transformation, modelling, and analysis
- 3_Cleaned_Data - Processed and structured dataset:
 - dim_customers_clean
 - dim_date_clean
 - dim_products_clean
 - dim_region_clean
 - fact_sales_clean
- 4_PowerBI - Dashboard files and visualizations
- 5_README.md - Project summary for GitHub
- 6_Ecommerce_Sales_Analysis.pdf – Detailed project documentation

Data Cleaning & Transformation (SQL)

Data Cleaning:

The raw dataset (`Sales\_data`) was initially explored and cleaned using SQL to ensure data quality and consistency.

The following steps were performed:

- Verified total number of records and previewed the dataset using `COUNT (\*)` and `TOP` queries
- Checked for missing values in key columns such as Profit, Sales, and Quantity
- Identified duplicate records based on Order_ID and product -level details

To remove duplicates, a `ROW NUMBER ()` window function was used:

- Partitioned by Order_ID, Product_ID, Sales, Quantity, Discount, and Profit
- Retained only unique records (rn=1)
- Created a cleaned table: `sales\_data\_clean`

Data standardization was applied:

- Removed leading/trailing spaces using `LTRIM ()` and `RTRIM ()`
- Standardized text formatting (e.g., City, State, Category) using proper capitalization

Data validation checks included:

- Negative values in sales and quantity
- Invalid discount values (outside 0-1 range)
- Incorrect date relationships (Order_Date > ship_Date)
- Null values in critical columns

Invalid records were handled by removing rows with null profit values.

The cleaned dataset was stored in `sales\_data\_clean` for further processing.

Data Modelling (Star Schema)

After cleaning, a star schema data model was created to support efficient analysis and reporting.

The model consists of one Fact table and multiple Dimension tables.

Dimension: Customers

The `Dim\_Customers` table was created to store unique customer information.

- Extracted distinct Customer_ID and Customer_Name
- Remove duplicates using DISTINCT
- Defined Customer_ID as the primary key

This table enables customer-level analysis.

Dimension: Products

The `Dim\_Products` table contains product-related information.

- Included Product_ID, Product_Name, Category, and Sub_Category
- Duplicate Product_IDs were resolved using GROUP BY and aggregation (MAX function)
- Defined Product_ID as the Primary key

This ensures each product is uniquely represented.

Dimension: Region

The `Dim\_Region` table stores geographical information.

- Includes Region, City, State, Country, and Postal_code
- Created a surrogate key: Region_ID using IDENTITY
- Set Region_ID as the Primary Key

This supports region-based analysis.

Dimension: Date

The `Dim\_Date` table was created to support time-based analysis.

- Extracted distinct Order_Date values
- Generated Date_ID as a surrogate key
- Derived additional attributes:
 - Year
 - Month
 - Month Name
 - Quarter

This enables trend and time-series analysis.

Fact Table: Sales

The `Fact\_Sales` table stores transactional data and connects all dimensions.

It includes:

- Order_ID
- Customer_ID
- Product_ID
- Region_ID
- Order_Date_ID
- Ship_Date_ID
- Sales, Profit, Quantity, Discount

The table was created by joining:

- Dim_Customers
- Dim_Products
- Dim_Region
- Dim_Date (for both order and ship dates)

A surrogate key (Sales_ID) was added as the Primary key.

Foreign key relationships were established with all dimension tables to maintain referential integrity.

Validation Checks

Several validation checks were performed to ensure data integrity:

- Verified foreign key constraints on all dimension and fact tables
- Checked foreign key relationships between Fact and Dimension tables
- Ensured no null values exist in key columns
- Compared row counts between cleaned data and fact table
- Identify duplicate records in Fact_Sales using GROUP BY

System tables (`sys. Indexes` , `sys. foreign\_keys`) were used to validate constraints.

These checks ensured the data model is reliable and consistent.

SQL Analysis (Business Insights)

SQL was used to perform business analysis and extract insights from the data.

Key analysis include:

- Total Sales and Order Volume
- Sales by Category and Region
- Monthly Sales Trends
- Top Performing Products
- Customer Purchase Behaviour

Advanced SQL techniques used:

- Common Table Expression (CTE)
- Window Functions (RANK, ROW_NUMBER, LAG)
- CASE statements for segmentation
- Aggregations and grouping

Additional insights generated:

- Sales segmentation (High, Medium, Low value orders)
- Monthly growth percentage analysis
- Repeat vs new customer identification
- Average order value calculation
- Shipping delay analysis
- Category-wise contribution to total sales

These analyses provide a comprehensive understanding of business performance.

The final dimension and fact tables were exported and stored in the `3\_Cleaned\_Data` folder for further use in reporting and dashboard development.

Power BI Dashboards

The Power BI dashboard was developed to provide an interactive and comprehensive view of E-commerce sales performance.

It consists of multiple report pages, each focusing on a specific business area:

- Executive Overview
- Sales Analysis
- Customer Analysis
- Business Insights & Strategic Recommendations

The dashboard includes KPIs, visualizations, filters, and dynamic measures to enable data-driven decision-making.

Key KPIs & Measures

The following key performance indicators (KPIs) and DAX measures were created:

- Total Revenue
- Total Profit
- Total Orders
- Total Customers
- Average Order Value (AOV)
- Average Revenue per Customer
- Profit Margin (%)
- Growth % (Month- over-Month)
- Repeat Customer & Repeat Customer %
- New Customers
- Revenue per Customer

Advanced Measures:

- Customer Segmentation (High, Medium, Low Value)
- Customer Contribution %
- Customer Ranking
- Top Category & Top Region
- Dynamic Titles and Dynamic Insights

These measures provide a comprehensive view of business performance.

Page 1: Executive Overview Dashboard

The Executive Overview dashboard provides a high-level summary of performance.

Key Elements:

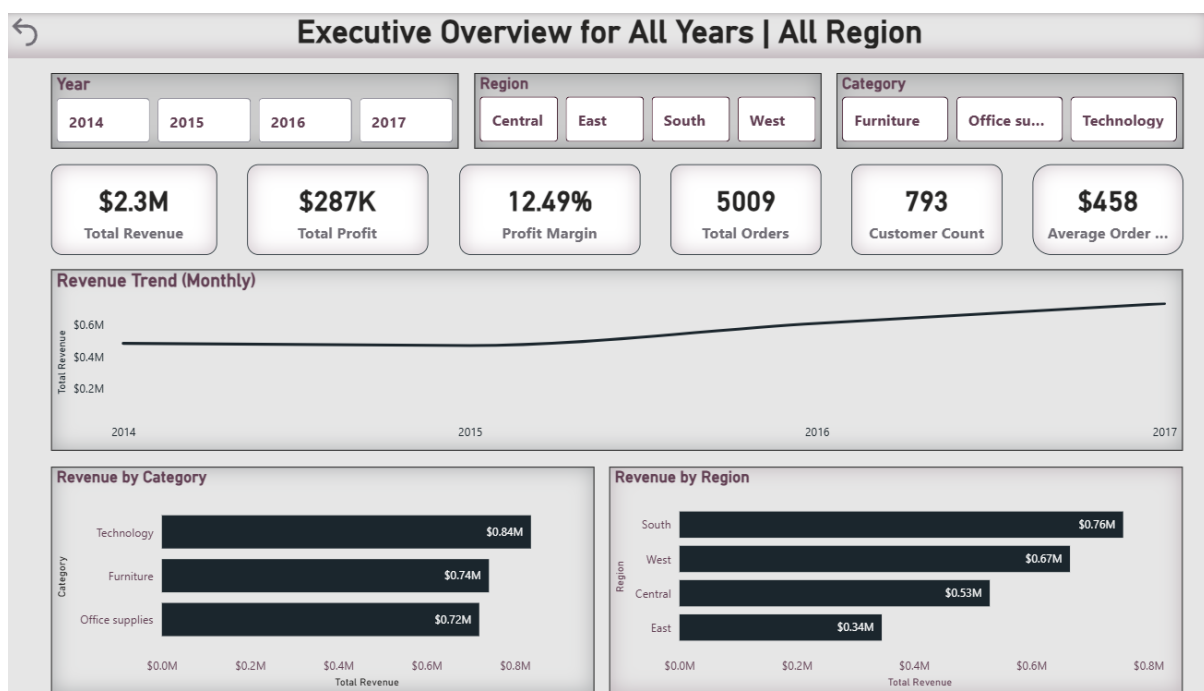
- KPI Cards: Revenue, Profit, Profit Margin, Orders, Customers, AOV
- Monthly Revenue Trend (Line Chart)
- Revenue by Category (Bar Chart)
- Revenue by Region (Bar Chart)

Filters/Slicers:

- Year
- Region
- Category

This dashboard allows stakeholders to quickly understand overall performance and trends.

Figure: Executive Overview Dashboard



Page 2: Sales Analysis Dashboard

This dashboard focuses on detailed sales performance analysis.

Key Visuals:

- Revenue by Category (Bar Chart)
- Category Contribution % (Donut Chart)
- Region Sales Performance (Bar Chart)
- Revenue & Growth Trend (Combined Chart)

Insights Features:

- Identification of top-performing categories and regions
- Analysis of revenue growth trends over time

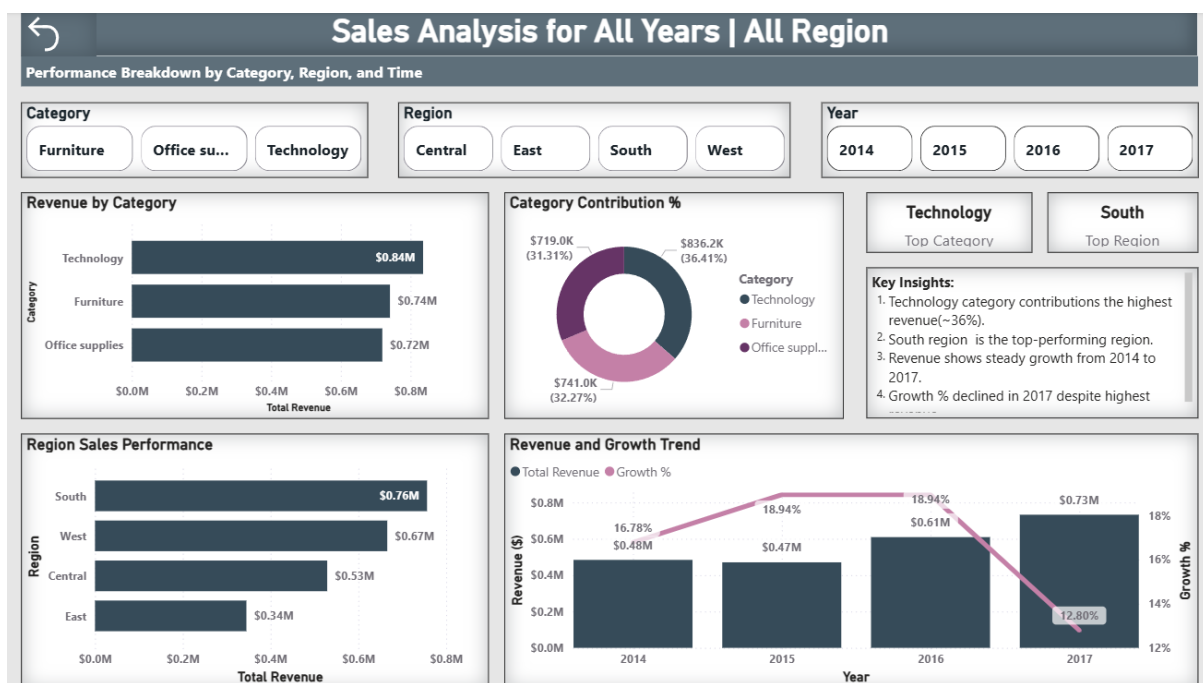
Dynamic highlights:

- Top Category
- Top Region

Filters/Slicers:

- Category
- Region
- Year

Figure: Sales Analysis Dashboard



Page 3: Customer Analysis Dashboard

This dashboard provides detailed analysis of customer behaviour.

Key KPIs:

- Total Customers
- Total Orders
- Avg Revenue per Customer
- Repeat Customers

Key Visuals:

- Top 10 Customers by Revenue (Table)
- Customer Distribution (Repeat vs New)
- Customer Segmentation (High, Medium, Low Value)
- Revenue Contribution by Customer Segment
- Customer Purchase Trend

Customer Segmentation:

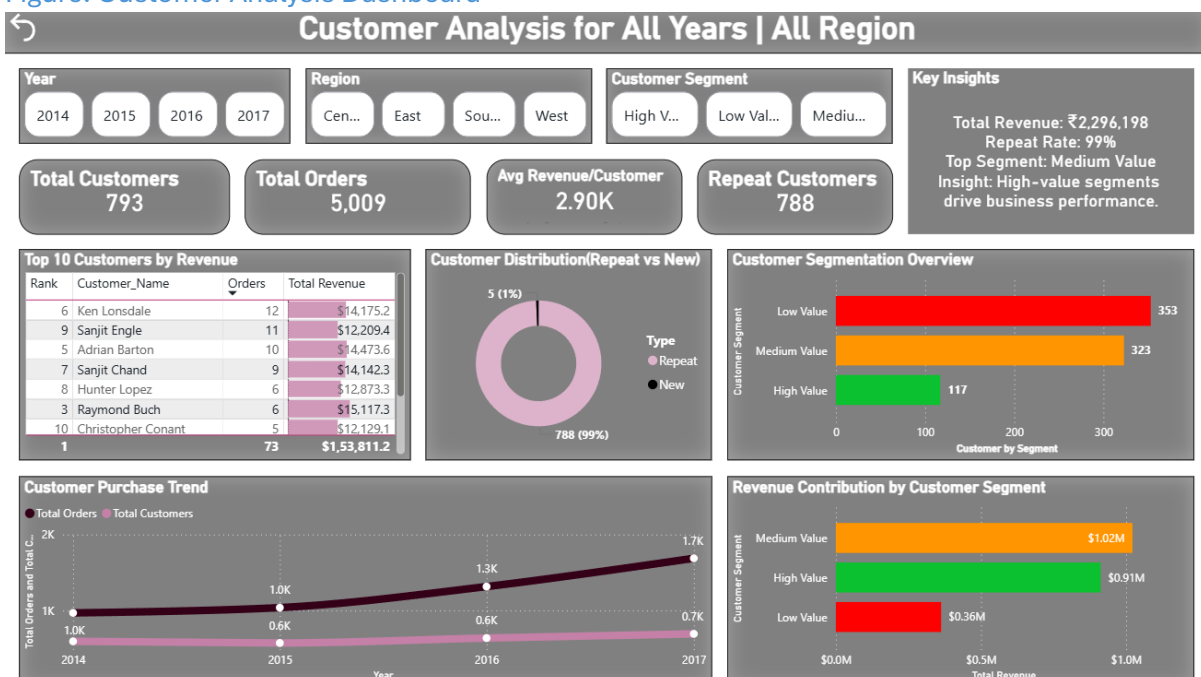
- High Value
- Medium Value
- Low Value

This segmentation was created in the Dim_Customers table and used for business analysis.

Filters/Slicers:

- Year
- Region
- Customer Segment

Figure: Customer Analysis Dashboard



Business Insights & Strategic Recommendations Dashboard

This dashboard presents key business finding and strategic recommendations.

Key Sections:

- Business Impact Summary
- Key Insights
- Recommendations

This page focuses on transforming data insights into actionable business strategies.

Figure: Business Insights & strategic Recommendations Dashboard



Interactivity Features

The dashboard includes interactive elements to enhance usability:

- Slicers for Year, Region, Category, and Customer Segment
- Dynamic Titles based on filter selection
- Cross-filtering between visuals
- Tooltip pages for detailed insights

These features allow users to explore the data dynamically.

Key Insights

The analysis of E-commerce sales data revealed several important business insights:

1. Revenue Concentration:

A significant portion of revenue is generated from a small group of high-value customers, indicating strong customer concentration.

2. Customer Segmentation Impact:

High-Value customers contribute the majority of total revenue, while low-value customers represent a large portion of total customers but contribute less revenue.

3. Category Performance:

Certain product categories consistently outperform others in both revenue and profitability, while some categories underperform and require attention.

4. Regional Variations:

Sales performance varies significantly across regions, with some regions contributing higher revenue and profit margins than others.

5. Sales Trend Analysis:

Revenue trends show fluctuations across months, indicating seasonal patterns and potential demand cycles.

6. Repeat Customer Behaviour:

Repeat customers contribute a substantial portion of revenue, highlighting the importance of customer retention strategies.

7. Profitability Insights:

Higher revenue does not always correspond to higher profit, indicating cost inefficiencies or discount-heavy sales in certain segments.

8. Growth Opportunities:

Certain customer segments and regions show growth potential but are currently underutilized.

Recommendations

Based on the insights derived from the analysis, the following recommendations are proposed:

1. Focus on High-Value Customers:

Implement loyalty programs, personalized offers, and premium services to retain and expand high-value customers.

2. Improve Low-Value Customer Conversion:

Develop targeted marketing strategies to convert low-value customers into medium or high-value customers.

3. Optimize Underperforming Categories:

Review pricing, marketing, and product strategies for low-performing categories to improve their contribution.

4. Regional Growth Strategy:

Invest in marketing and logistics improvements in underperforming regions to increase sales performance.

5. Enhance Customer Retention:

Introduce retention strategies such as discounts, memberships, and engagement campaigns to increase repeat purchases.

6. Profit Optimization:

Analyse cost structures and discount strategies to ensure higher profitability across all categories.

7. Seasonal Planning:

Leverage seasonal trends by planning promotions and inventory based on historical sales patterns.

8. Data-Driven Decision Making:

Continuously monitor KPIs and update dashboards to support real-time business decisions.

Conclusion

This project demonstrates the end-to-end implementation of a Business Intelligence solution using SQL and Power BI.

The workflow included:

- Data cleaning and transformation using SQL
- Data modelling using dimension and fact tables
- Data validation and business analysis
- Interactive dashboard development in Power BI

The final solution provides actionable insights into sales performance, customer behaviour, and business growth opportunities.

This project highlights the ability to:

- Work with structured datasets
- Apply business logic to data
- Build interactive dashboards
- Translate data into meaningful business insights

Overall, this project reflects strong analytical thinking and the ability to support data-driven decision-making in a real-world business scenario.